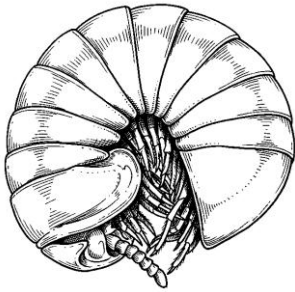


Millipedes- Diplopoda

roll into perfect sphere.



hard body cover,
roll into spiral



soft body, tufts of
setae in tergite



Helminthomorpha

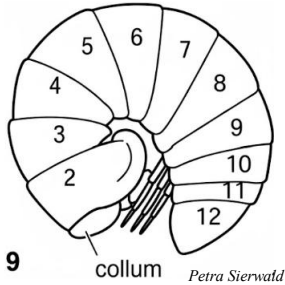
2

Polyxenida

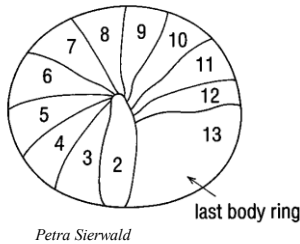
1

12 segments

13 segments



Glomerida



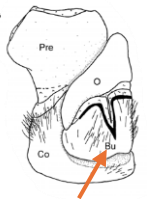
Sphaerotheriida

No fusion...other families **NR**

Fusion of two basal
plates of vulva into single
plate: Zephronidae



Holloway 1956



Wesener 2015

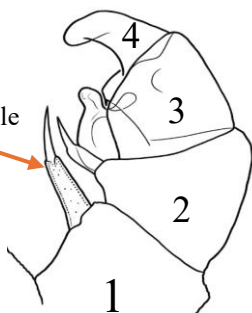
One pair of telopods (Glomeroidea);

Lamellar processes on medial face of
telopod femur and tibia; accompanied
by setae; finger shaped process on
prefemur (Glomeridae);

Telopoditomere 1-3 of telopod with
trichostyles; 2 and 3 with large
processes

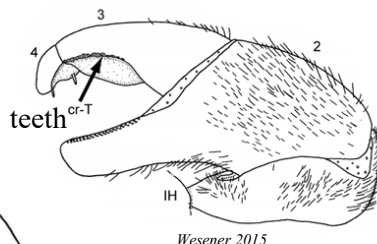
Hyleoglomeris

trichostyle



Golovatch 1987b:8

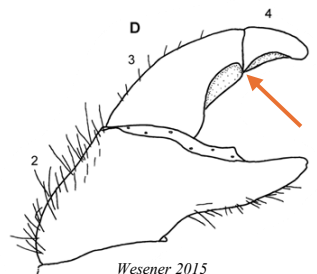
Presence of crenulated
sclerotized teeth on podomere 3
or 4 and membranous lobe on
podomere 2 of posterior
telopods; crenulated teeth
present on podomere 2 of
anterior telopods



Wesener 2015

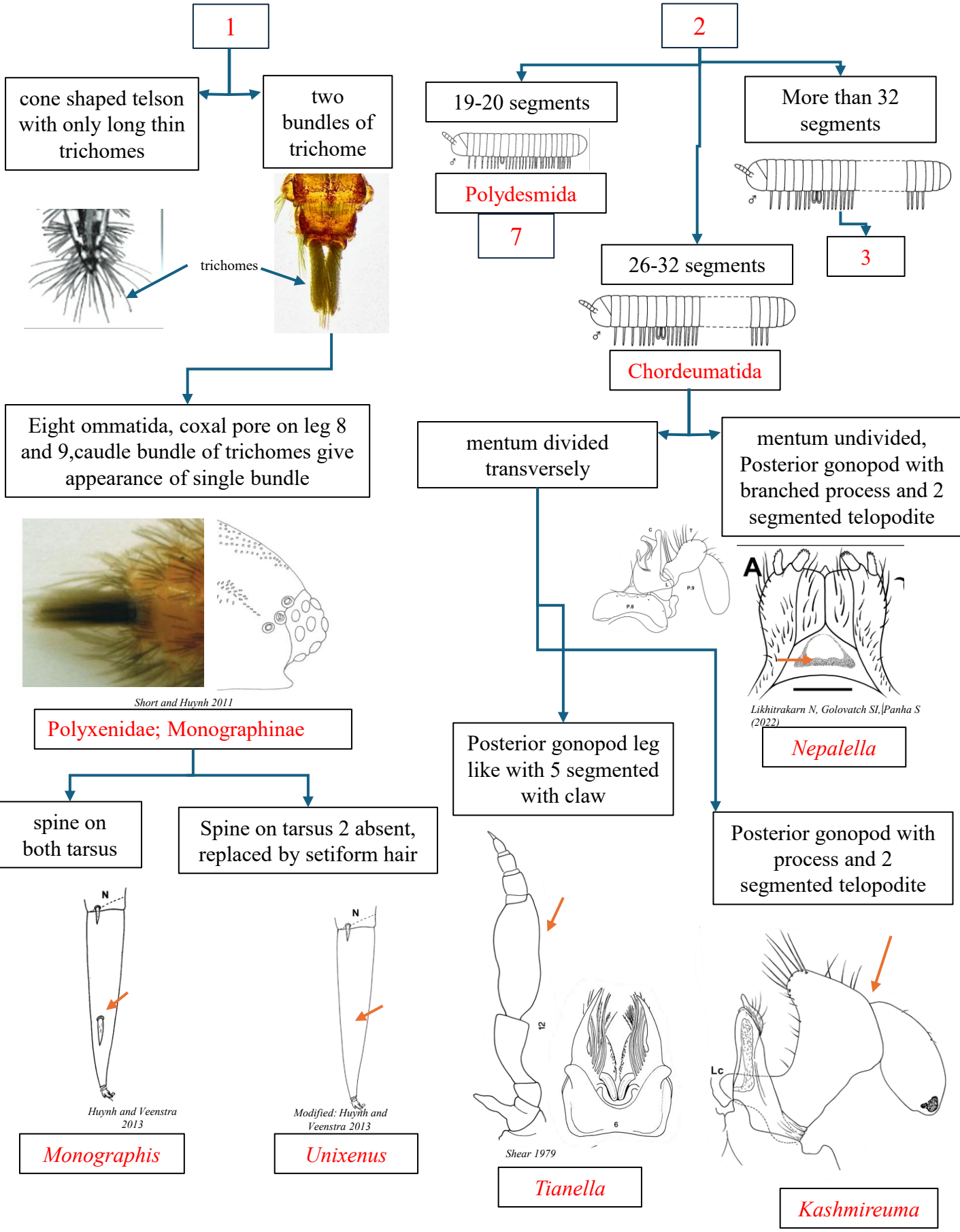
Zephronia

Absence of crenulated
sclerotized teeth on
podomere 3 or 4 and
membranous lobe on
podomere 2 of
posterior telopods;
crenulated teeth absent
on podomere 2 of
anterior telopods

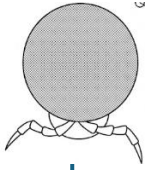
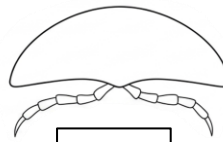


Wesener 2015

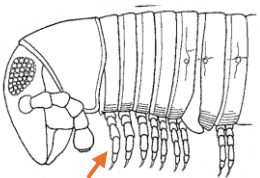
Kophosphaera



3

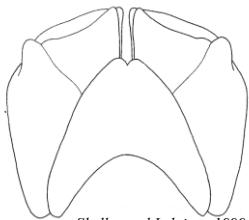
Cross section of
ring circularCross section of
ring flattened

5

First pair legs same
as other legs

Shelley and Lehtinen 1999

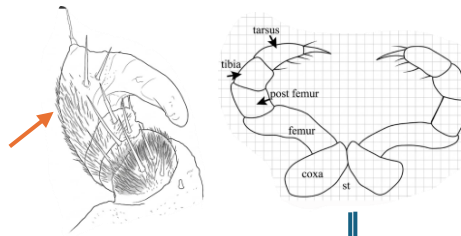
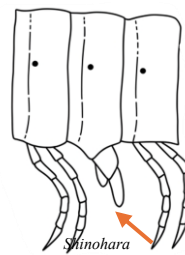
Posterior gonopod
connected by
sclerotized sternite;
Anterior gonopod box-
like, coxite and
telopodite reaching
similar height, lacking
any processes



Shelley and Lehtinen 1999

Trigonius

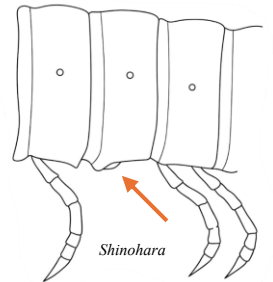
If no sternite
and some
different
characters...
new family,
new genus for
Nepal

First Pair of legs different
than others remaining legsGonopod clearly visible
from lateral side

Shinohara

Julida

first pair of leg modified
into hook shaped and
smaller than subsequent
legs: Julidae

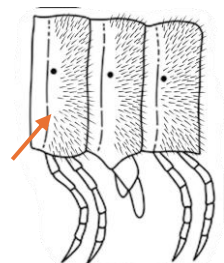
Gonopod not visible
from lateral side

Shinohara

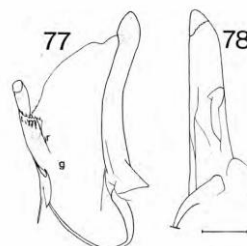
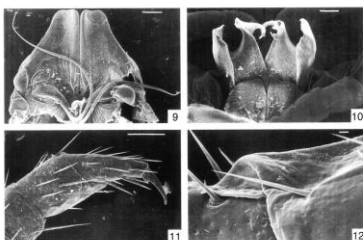
Spirostreptida

4

Metazonite without setae

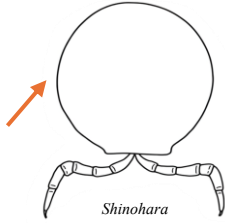
*Anaulaciulus*

Metazonite with setae

*Nepalmatoiulus*

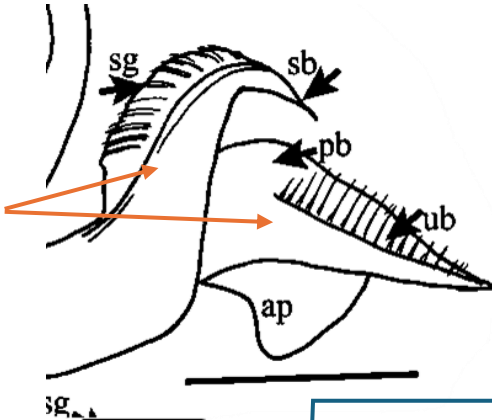
4

Metatergite smooth

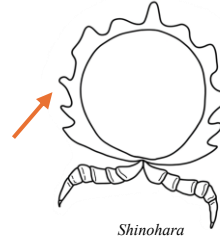


Harpagophoridae

Gonopod distal telopodite with two branch: seminal and pectinate branch

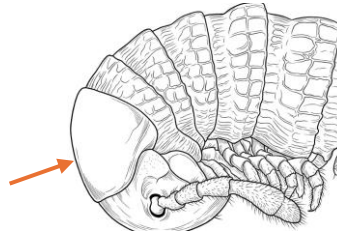


Metatergite with crest



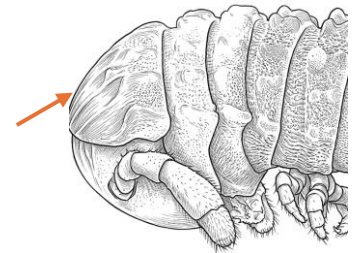
Cambalopsidae

Collum smooth, without strong longitudinal crests



Trachyjulus

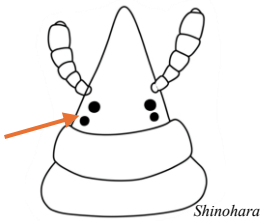
Collum with longitudinal crests



Podoglyphiulus

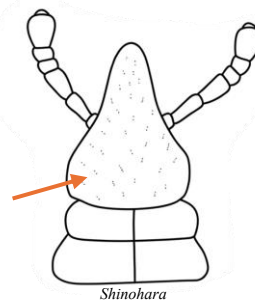
5

eyes present

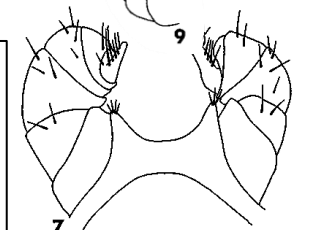
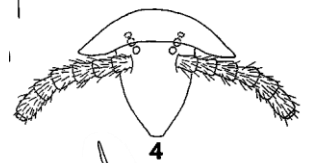


Polyzoniida

No eyes



6



Nepalozonium

Posterior

Shelley 1996

Posterior tergal margin upturned. Elevated above succeeding tergites

Hirudisomatidae

Three eyes, Anterior and posterior gonopod with six podomeres, anterior gonopod sternite with two setose lobes. Podomere 6 of posterior gonopod slender and strongly elongated.

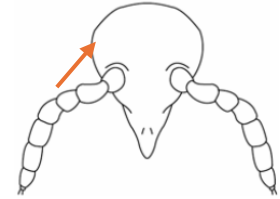
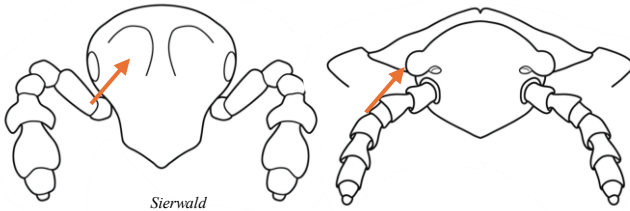
6

If beak, Siphonophorida; **SIPHONOPHORIDAE**

Head pear shaped,
subtriangular, not
extended into beak

Head with bulges
above antenna
socket, with distinct
lateral paranota;
more than 35 rings;

No bulges above
antenna socket, with
distinct lateral
paranota; more than
35 rings;



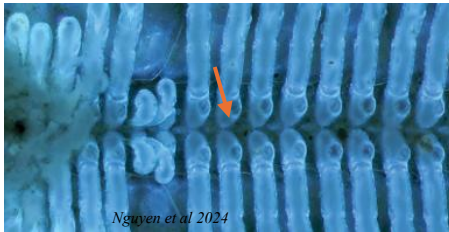
Platydesmida

**Siphonophorida;
Siphonorhinidae**

Sterna narrow, coxae
almost touched medially

Sterna very wide, coxae of
leg well separated from
one another

10-50 mm long. Coxae 1
fused with the sternite.
Apical podomere of
posterior gonopod divided
into two branches lacking
spines



Platydesmidae, NR

Andrognathidae

wide trunk, strongly elongated paranota,
uoto 60mm long, 66-76 segments, collum
with tubedcles, tergite densely pilose,
strongy sculptured with tubercles, one
row of larger tubercles



Modified Anilkumar et al. 2026

Siphonorhinus



Modified Nguyen et al 2024

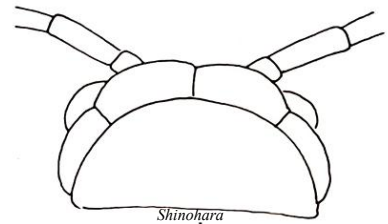
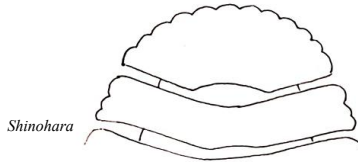
Pseudodesmus

*NR= Not Recorded (Just
mentioned it to
differentiate) in all pages*

7

Head can't seen from above

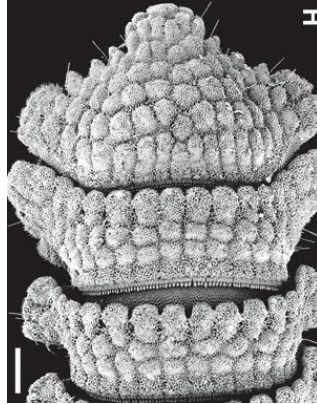
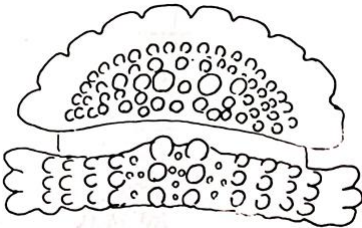
Head seen from above



8

Tergite flat, radiating lines and lobulations in collum, tubercles in metatergite, paraterga with marginal lobulation

convex subcylindrical shaped tergite



Golovatch and VANDEN SPIEGEL 2014

Pyrghodesmidae

Haplodesmidae

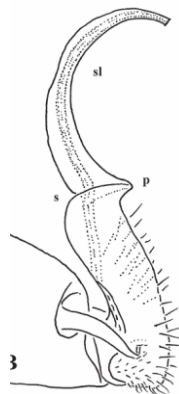
fore margin of collum is convex with 6+6 subequal sectors divided by 11 radii, 3+3 and 2+2 tubercles, paraterga 2 and 3 are trilobate then following 2, 4 or 5 lobulate in lateral margin;

19 segments, ozopore formula ((5,7-18), 4 transverse setigerous isostictic tubercles, clearly helicoid, twisted prefemoral portion of the gonopod



Nyugen.

Cryptocorypha



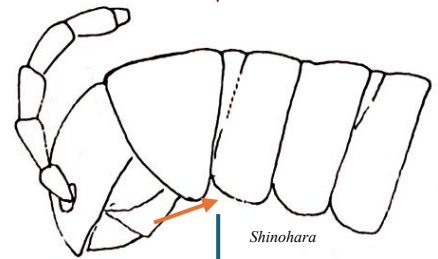
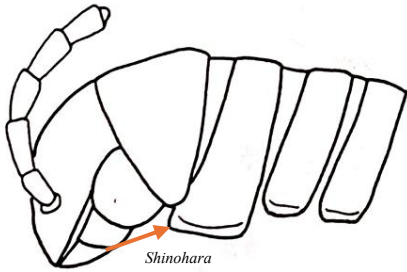
Golovatch and VANDEN SPIEGEL 2014

Koponeius

8

Tergite 2 lower than
collum, lateral view

Tergite and collum in same
level, lateral view

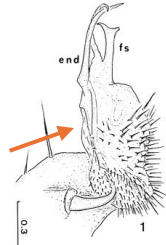


Paradoxosomatidae

9

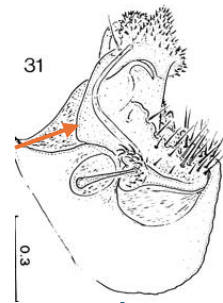
Large shaped coxae but
femorite free, not sunken
in gonocoel

Large shaped coxae but
femorite deep sunken
into gonocoel



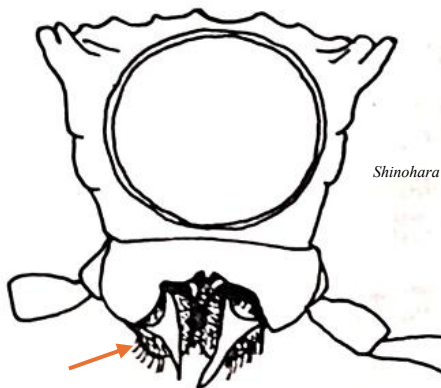
Polydesmidae

12



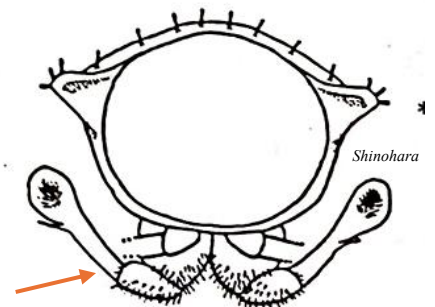
Gonopod teleopodite directed
forward or backward

Telopodite facing laterally;
teleopodite suberect, no process at
the base of femotire



Trichopolydesmidae

10



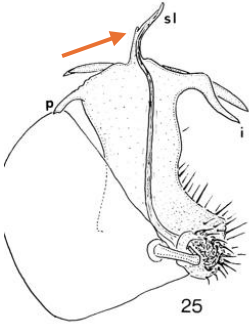
Opisotretidae

11

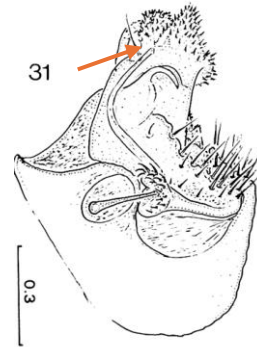
10

Solenomerite log

Solenomerite short



Hingstonia



Solaphilus

11

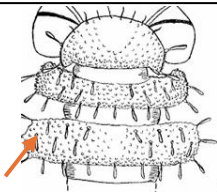


Martensodesmus

12

Tergal sculpture absent; beak shaped paraterga posterior part; seminal branch coiled; femurite end like blade and spine like hairs

Tergal sculpture present

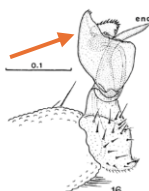


Glenniea

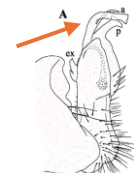


Premorite is simple, gonofemoral process prominent

Prefemorite part strongly expanded; gonofemoral processes thin



Himalodemus



Epanerchodus

9

solenophore not divided into lamina
lateralis and lamina medialis

solenophore not divided into lamina
lateralis and lamina medialis

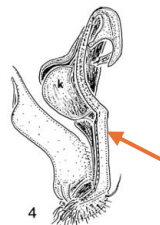
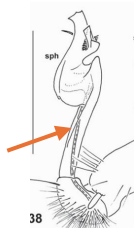
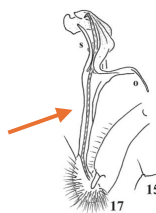
Alogolykinae

Paradoxosomatinae

13

Elongated femorite

Femorite shortened

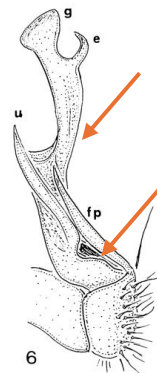
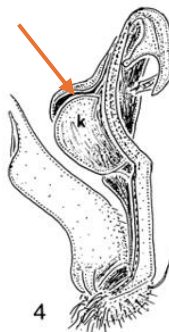
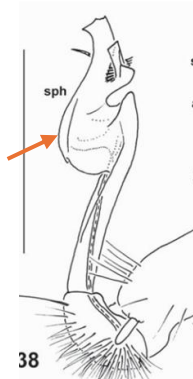
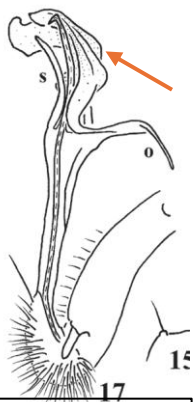


Leaf shaped
tibiotarsus, spine at
the end of femorite

Boat shaped
tibiotarsus

femorite
longer than
post
femoral
part, post
femoral part
broad

femorite very
short, post
femoral part
long, rod shaped,
solenomere long

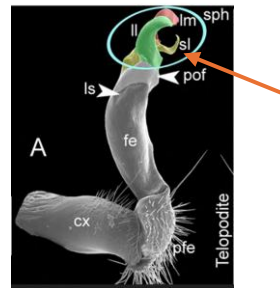


Delarthum

Pocockina

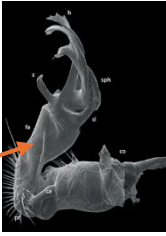
Hirtodrepanum

Tourenella

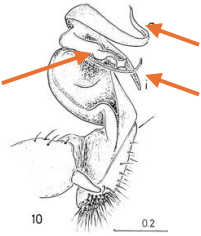
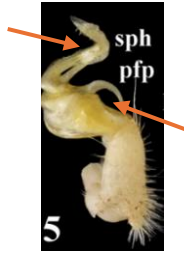
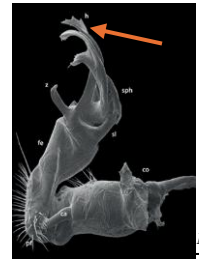


13

femorite grooved mesally

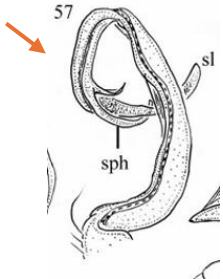


Sulciferini

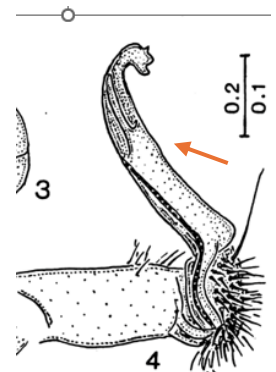
solenophore cover
solemnomere, 2
more processes*Anoplodesmus*Basal one process;
long solenophore
covering
solenomere*Chondromorpha*femorite distally
have fork shaped
processes*Kronopolites*solenophore cover
solenomere, 2
extra processes,
solenophore tip
have two tip*Oxidus*

NGUYEN et al 2017

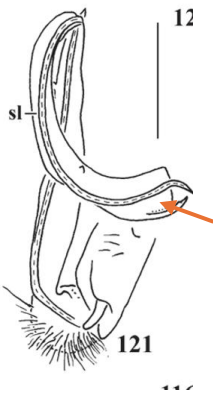
14

femorite long and
slender; post femoral
portion long, slender,
coiled, thin solenomere
and complex
solenophore.....Cha
mberlini, 16*Chamberlini*

15

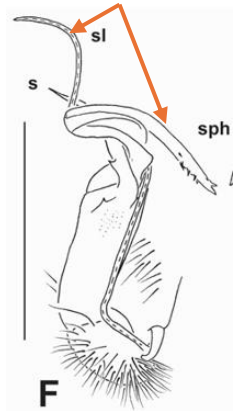
femorite long slender,
modestly curved; elongated
tibiotarsal end, no process at
the end of femorite,
solenophore curved mesad or
caudo mesad and consist
laminae lateralis and medialis,
sheath solenomere.*Orthomorpha*
*Orthomorpha*process at the end of
femorite; thick
solenomere gradually
continuing femorites
outline.....
Paradoxosomatini,*Substrongylosoma*

femorite end with two processes, long solonomere supported by equal length ribbon shaped solenophore



Beronodesmus

no process on femorite before sulcus, ribbon shaped solenophore with parabasal process; independent solonomere



Beronodesmoides